

DBMS Unit 2 — Relational Model & SQL Complete Notes

Database Management Systems (DBMS) · Unit 2 · Relational Model & SQL · 2025

Overview

These notes cover **Relational Model & SQL**. Design, query and manage relational databases — from ER modelling and SQL to normalization, transactions and recovery. Each section states the core idea, a worked intuition and the points examiners look for.

1. Relational Algebra

Selection, projection, union, set difference, Cartesian product and rename form the procedural query language behind SQL.

Why it matters: Relational Algebra appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of relational algebra in one line.
- A small example showing relational algebra in practice.
- Common mistakes students make and how to avoid them.

2. Joins

Inner, natural, left, right and full outer joins combine tuples from relations based on a join predicate.

Why it matters: Joins appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of joins in one line.
- A small example showing joins in practice.
- Common mistakes students make and how to avoid them.

3. Nested Queries

Subqueries with IN, EXISTS, ANY and ALL allow a query to use the result of another query, correlated or uncorrelated.

Why it matters: Nested Queries appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of nested queries in one line.
- A small example showing nested queries in practice.
- Common mistakes students make and how to avoid them.

4. Aggregate Functions

COUNT, SUM, AVG, MIN and MAX with GROUP BY and HAVING summarise groups of rows.

Why it matters: Aggregate Functions appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of aggregate functions in one line.
- A small example showing aggregate functions in practice.
- Common mistakes students make and how to avoid them.

5. Views

A view is a virtual relation defined by a query; updatable views must map unambiguously to base tables.

Why it matters: Views appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of views in one line.
- A small example showing views in practice.
- Common mistakes students make and how to avoid them.

6. Integrity Constraints

NOT NULL, UNIQUE, CHECK, PRIMARY KEY and FOREIGN KEY constraints protect data correctness.

Why it matters: Integrity Constraints appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of integrity constraints in one line.
- A small example showing integrity constraints in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Relational Algebra · Joins · Nested Queries · Aggregate Functions · Views · Integrity Constraints